BTech

Chemistry

Quiz

Date: 4th Feb 2026

Total Marks = 25

Time = 35 min

Write your **full name, roll number and date** clearly on the first page of your answer sheets and any additional page you take

Questions should be clearly answered. No ambiguous answers will be marked (subjected to instructor's discretion)
For MCQs: Mention option along with the answer

NO ONE SHOULD CARRY MOBILE PHONES OR ANY NOTES WITH THEM DURING THE QUIZ

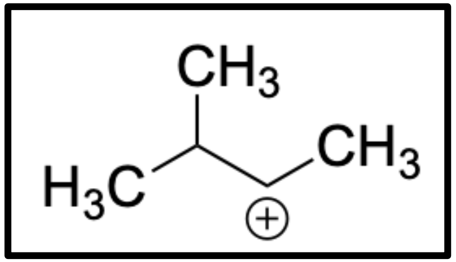
MCQs carry 1 Mark each – 15 MCQs

Answers on first page only

No explanations are required for any answer

Slides are clocked and will not be repeated on request

Q1. How many hyperconjugation structures are possible for the following carbocation?



(A) 4 (B) 3 (C) 9 (D) 6

Q2. Which of the following solvents is likely to dissolve longer chain fatty acids?

(A) Methanol (B) Water (C) Hexane (D) None of the above

Q3. Which of the following functional groups are not present/ encountered in human cells?

(A) Amides (B) Thiols (C) Epoxides (D) Azides

Q4. Which of the following aqueous mixture is a buffer solution?

(A) Acetic acid and Ammonium chloride (B) Hydrochloric acid and Potassium acetate (C) Acetic acid and Sodium Chloride (D) Acetic acid and Sodium acetate (E) Sodium hydroxide and Potassium Acetate

Q5. Which of the following reaction is applied for DNA sequencing in sequence-by-synthesis (SBS) method?

(A) Copper assisted Click Reaction (B) Cannizaro's reaction
(C) Staudinger's reaction and ligation (D) None of the above

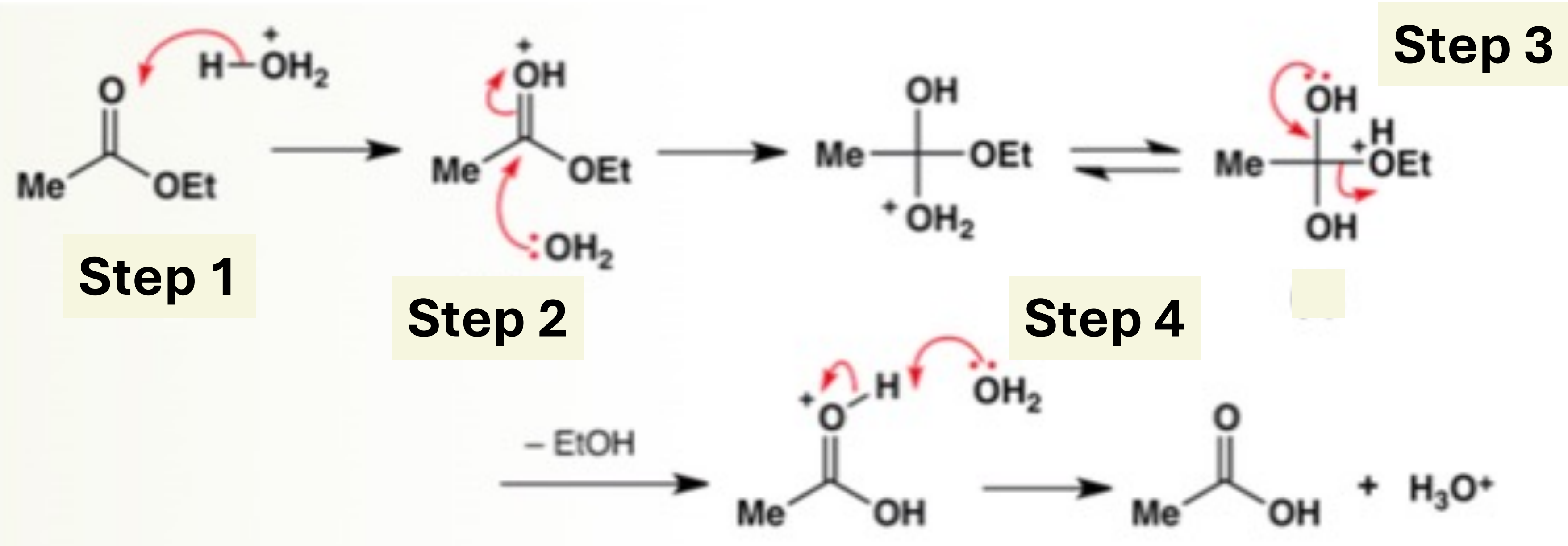
Q6. The dihedral angle in the most stable conformer of 1,2-difluoroethane is

(A) 0 degree (B) 120 degree (C) 60 degree (D) 180 degree

Q7. The most stable conformer of cyclohexane is?

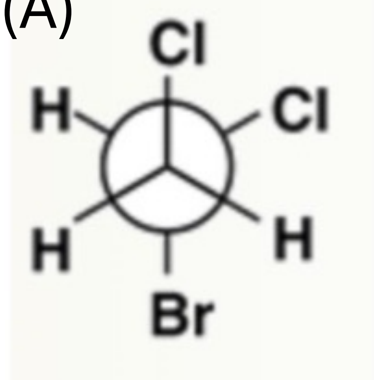
(A) Half-chair (B) Twist Boat (C) Chair (D) Boat (E) Flat

Q8. In the below reaction, which of the following steps has a wrong reaction mechanism (arrow pushing)?



(A) Step 1 (B) Step 2 (C) Step 3 (D) Step 4

Q9. The most stable conformer of 1-bromo-1,2-dichloroethane is?

(A)


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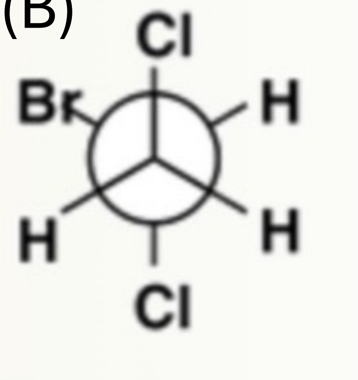

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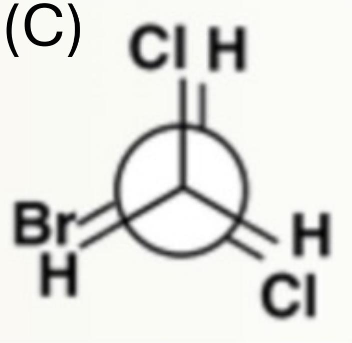
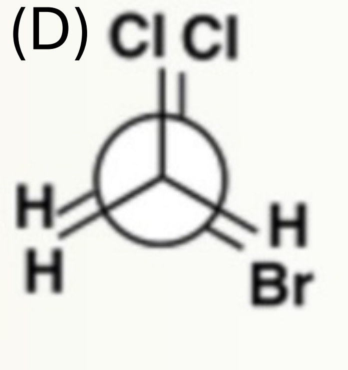


(D)


Q10. Rhodamines generally show fluorescence in the range

(A) 400-460 nm (B) 490-520 nm (C) 540-600 nm (D) 640-680 nm

Q11. Which of the following is a natural polymer?

(A) polybutadiene (B) poly (Butadiene-acrylonitrile)

(C) cis-1,4-polyisoprene (D) PEG 1000 KDa

Q12. Which of the following groups has the highest priority according to Cahn-Ingold-Prelog (CIP) sequence rules?

A) $-C\equiv CH$ B) $-CH=CH_2$ C) $-CH(OH)CH_3$ D) $-CH_2CH_2OH$

Q13. For designing chiral drugs, why is knowledge of molecular symmetry important?

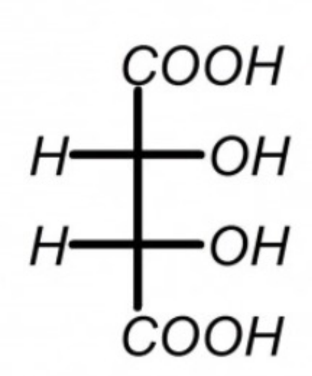
A) Because it always drastically affects the cost of synthesis
B) It helps predict chiral interactions, physical properties and spectroscopic behavior of drugs
C) It replaces the need for any biological testing
D) Both options A and B are correct
E) Knowledge of symmetry is not important

Q14. The instrument used to measure optical rotation of molecules is called?

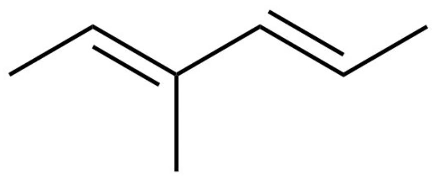
A) Infrared spectrophotometer B) Polarimeter

C) Optometer D) X-ray diffractometer

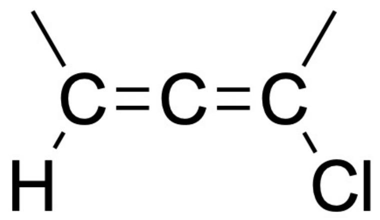
Q15. Which is the below molecules (A-D) has a traditional chiral center, but does not have a stereogenic center?



(A)



(B)



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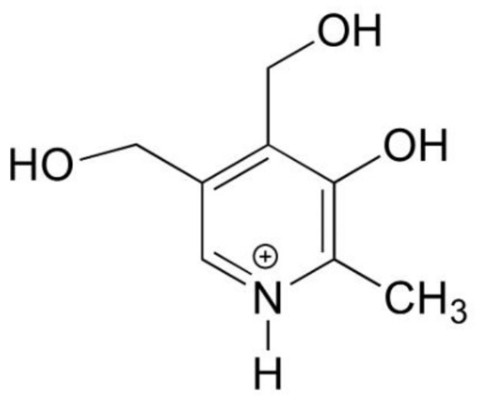
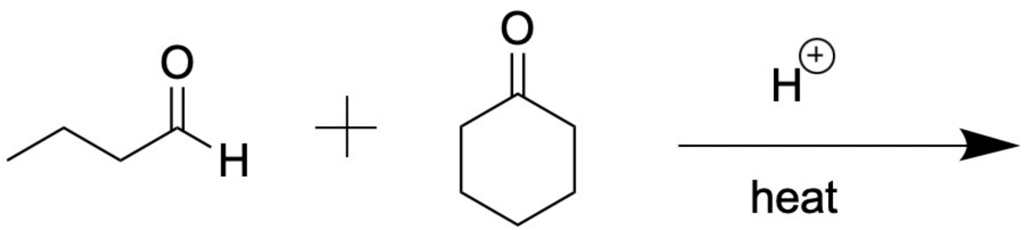


(D)

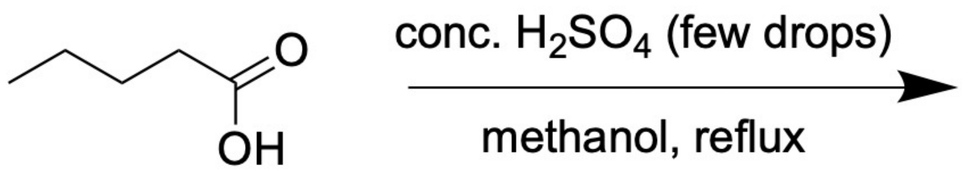
END of MCQs

Q16. Predict the **Major Product** for the following reactions:
(Mention possible stereoisomers, if any) [1x 3 = 3 Marks]

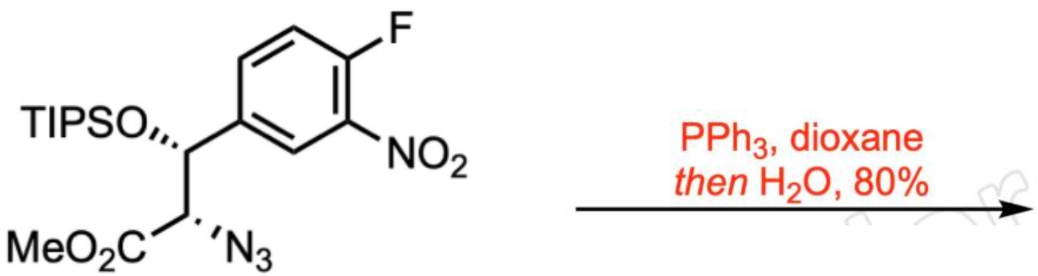
A.



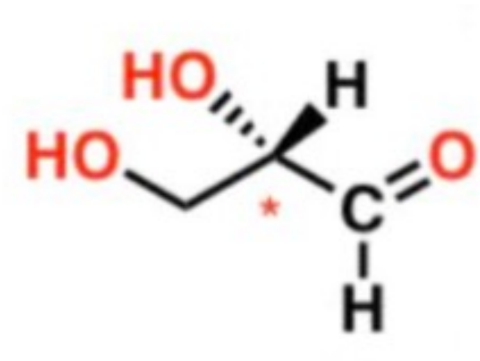
B.



C.



Q17. Assign D or L configuration to each of the following molecules (1x4 = 4 Marks)



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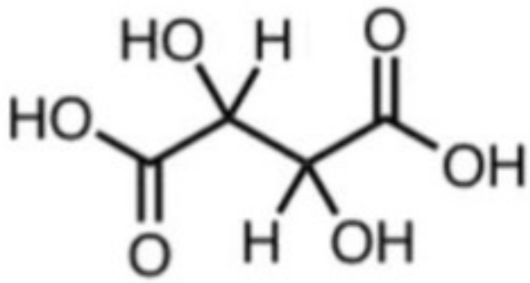

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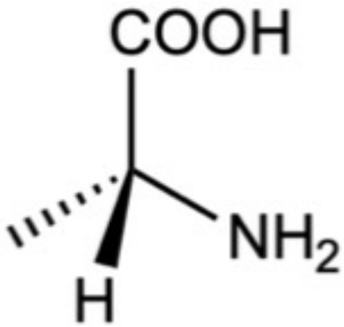

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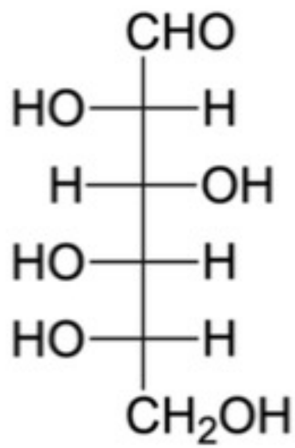

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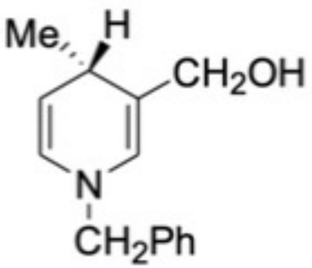
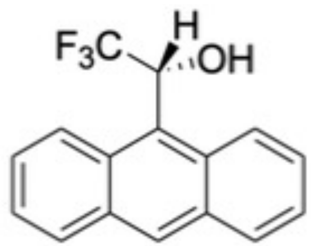
Q18. Assign R or S configuration to each of the following molecules (1x3 = 3 Marks)



1



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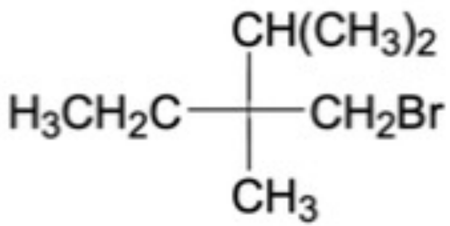

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END of QUIZ
Thank you!